Task 1 Report

Team: T26

Dept: Cs

احمد ايمن محمود خليفة عبد الله	2023170026
مي أحمد ماجد عبدالنعيم	2023170649
نهى يوسف محمد موسى	2023170669
ملك محمد محمد عزمي	2023170627
ندى محمد محمد عبد الباقي	2023170665
بسملة خالد عبد الخالق احمد	2023170137

Introduction

This project uses a penguin dataset containing samples from three species: Adelie, Chinstrap, and Gentoo, with 50 samples per species. Each sample has five features: CulmenLength, CulmenDepth, FlipperLength, OriginLocation, and BodyMass.

The dataset is used to compare the Perceptron and Adaline learning algorithms by selecting two features and two species for each experiment. For each selected species, 30 samples are randomly chosen for training and the remaining 20 for testing. The algorithms' performance is evaluated using confusion matrices, overall accuracy, and visualizations of the learned decision boundaries to examine how effectively different feature combinations distinguish between species.

Main Observations after modeling

There are some interesting observations when applying both algorithms on different feature combinations.

Easy pairs to differ

Adelie vs Gentoo

Firstly, separable species like Adelie vs Gentoo are easy. Almost every feature pair reaches 92.5-100% for both algorithms. Chinstrap vs Gentoo is easy too. Both algorithms are at 92.5% or above for every pair.

Hard Pairs to differ

Adelie vs Chinstrap

On the other hand, Adelie vs Chinstrap is the hard pair, and it is where the algorithms differ. It is the only pair that reaches 45-60% accuracy, which is chance level. CulmenLength + CulmenDepth is the only combination that gets 100% for Adelie vs Chinstrap. Every pair that uses CulmenDepth *without* CulmenLength collapses (CulmenDepth + BodyMass 50% / 47.5%, CulmenDepth + OriginLocation 45% / 50%). So CulmenDepth alone doesn't separate these two species, and CulmenLength is the key feature for this pair.

The two worst examples (4 & 5 presented below) show it clearly: FlipperLength + BodyMass gave 57.5% for the Perceptron against 85% for Adaline, and CulmenDepth + OriginLocation gave 50% against 67.5%. In the second one the Perceptron predicted Chinstrap for all 40 test samples.

Perceptron vs Adaline

Perceptron vs Adaline: on this hard pair Adaline is often higher (FlipperLength + BodyMass 77.5% vs 60%, CulmenDepth + FlipperLength 75% vs 57.5%, OriginLocation + BodyMass 72.5% vs 50%). But it isn't always (CulmenLength + FlipperLength 82.5% Perceptron vs 95% Adaline in the other direction is the opposite, and CulmenDepth + BodyMass is a tie at chance).

Another important Observation is in the OriginLocation feature:

OriginLocation is a special case. It gives 100% on Chinstrap vs Gentoo with every partner feature, including CulmenDepth and BodyMass. That is consistent with the two species living on different islands. For any other features it's nearly inseparable producing around a 50/50 chance of getting the species right, which means both of these species are fairly similar in every aspect. It's just their location that determines which species are which. This is a very interesting observation that helped us determine that we have to use the OriginLocation feature to distinguish between those 2 species. For Adelie vs Chinstrap it is weak, since Adelie lives on several islands.

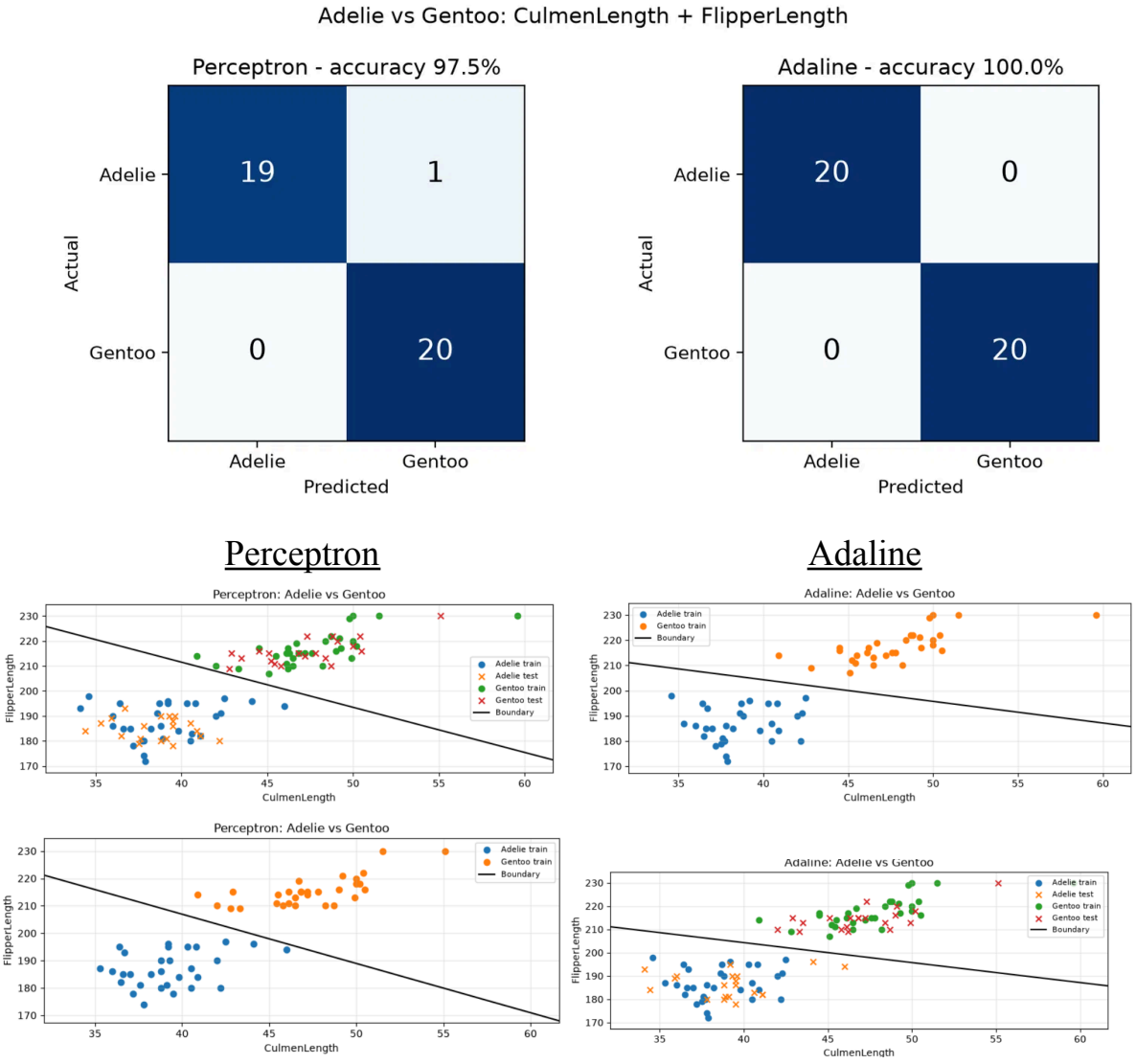
Main Informative Combinations

Test accuracy for every feature pair and class pair (single 30/20 split, learning rate = 0.01, 200 epochs).

Number	Classes	Features	Perceptron / Adaline	General Note
1	Adelie vs Gentoo	CulmenLength + FlipperLength	100% / 100%	Good, clearly separable
2	Chinstrap vs Gentoo	CulmenDepth + OriginLocation	100% / 100%	Good, shows the island effect
3	Adelie vs Chinstrap	CulmenLength + CulmenDepth	100% / 100%	Good, right features fix the hard pair
4	Adelie vs Chinstrap	FlipperLength + BodyMass	60% / 77.5%	Bad, overlap, Adaline more stable
5	Adelie vs Chinstrap	CulmenDepth + OriginLocation	45% / 50%	Worst case, near 50/50 chance

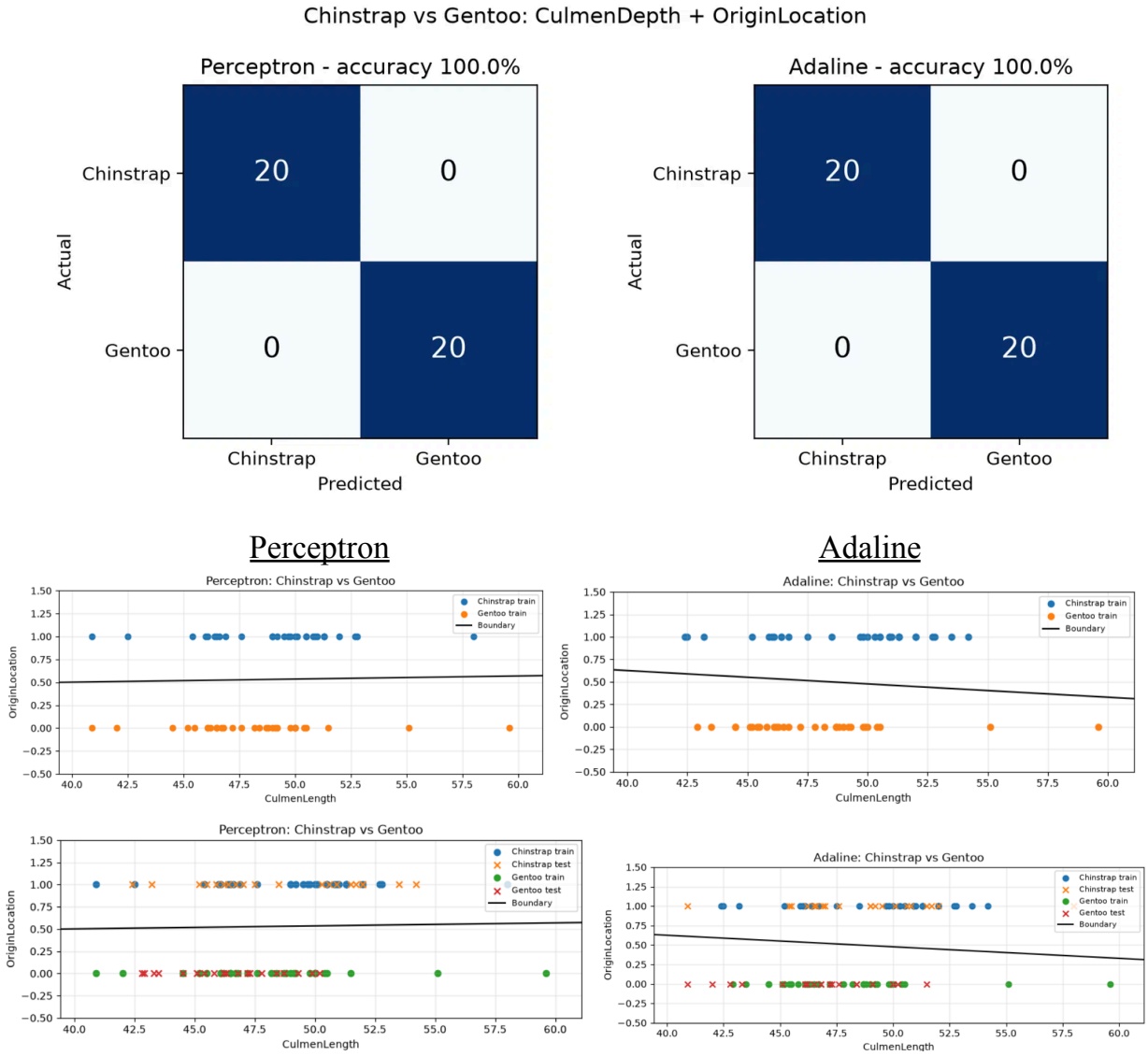
Detailed Analysis of the 5 Combinations

Adelie vs Gentoo -> CulmenLength + FlipperLength



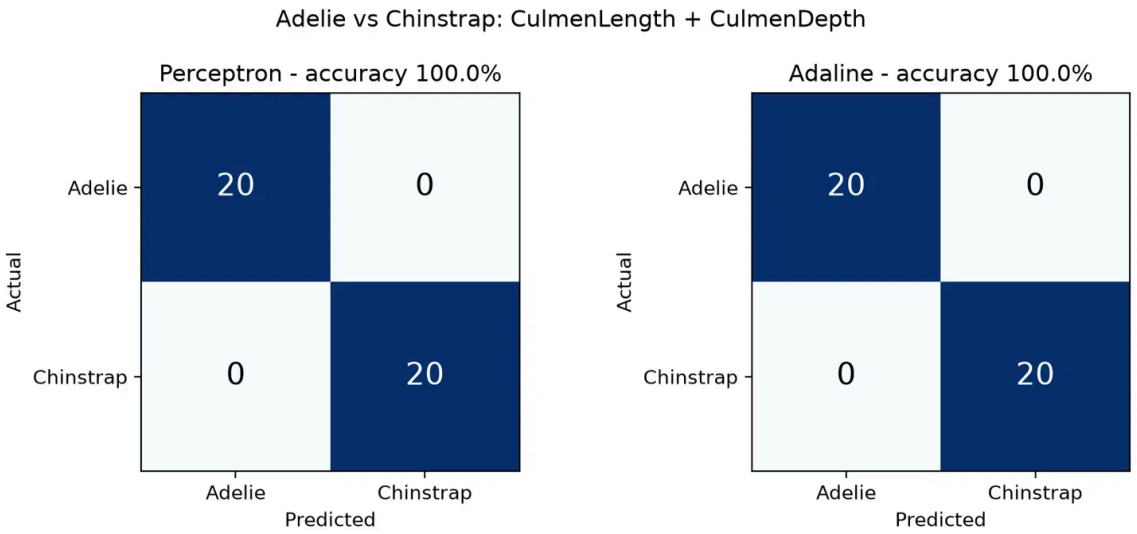
Both algorithms performed very well when distinguishing Adelie from Gentoo penguins. The Perceptron correctly classified 39 out of 40 samples, while Adaline achieved 100% accuracy. This strong performance is likely due to the clear differences in flipper and bill lengths between the species, making them easy to separate using a linear decision boundary. The small difference in results likely reflects the random train-test split rather than a fundamental weakness in either algorithm.

Chinstrap vs Gentoo -> CulmenDepth + OriginLocation

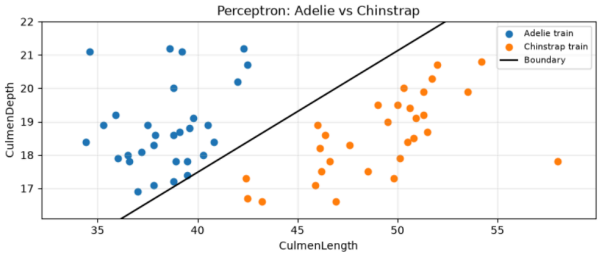


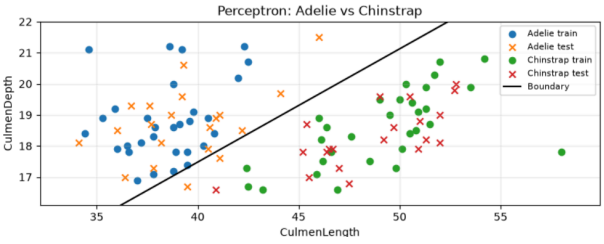
Both algorithms achieved 100% accuracy when classifying Chinstrap and Gentoo penguins. Culmen depth helps distinguish the species, while OriginLocation further improves separation because they were recorded on different islands (as mentioned previously in the observation after modeling). However, this perfect performance partly reflects geographical differences rather than physical characteristics alone, so OriginLocation should be interpreted with caution.

Adelie vs Chinstrap - CulmenLength + CulmenDepth

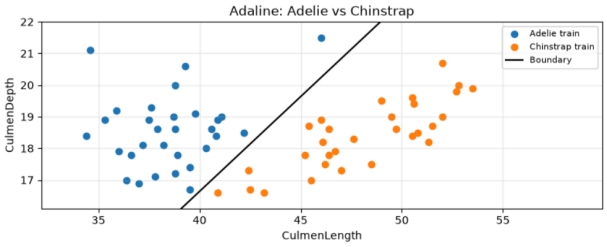


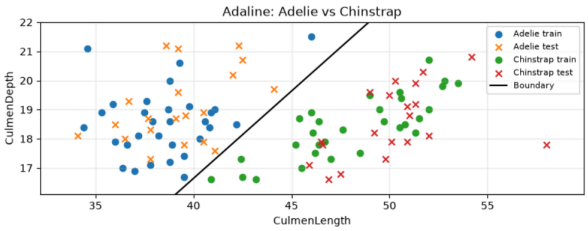
Perceptron





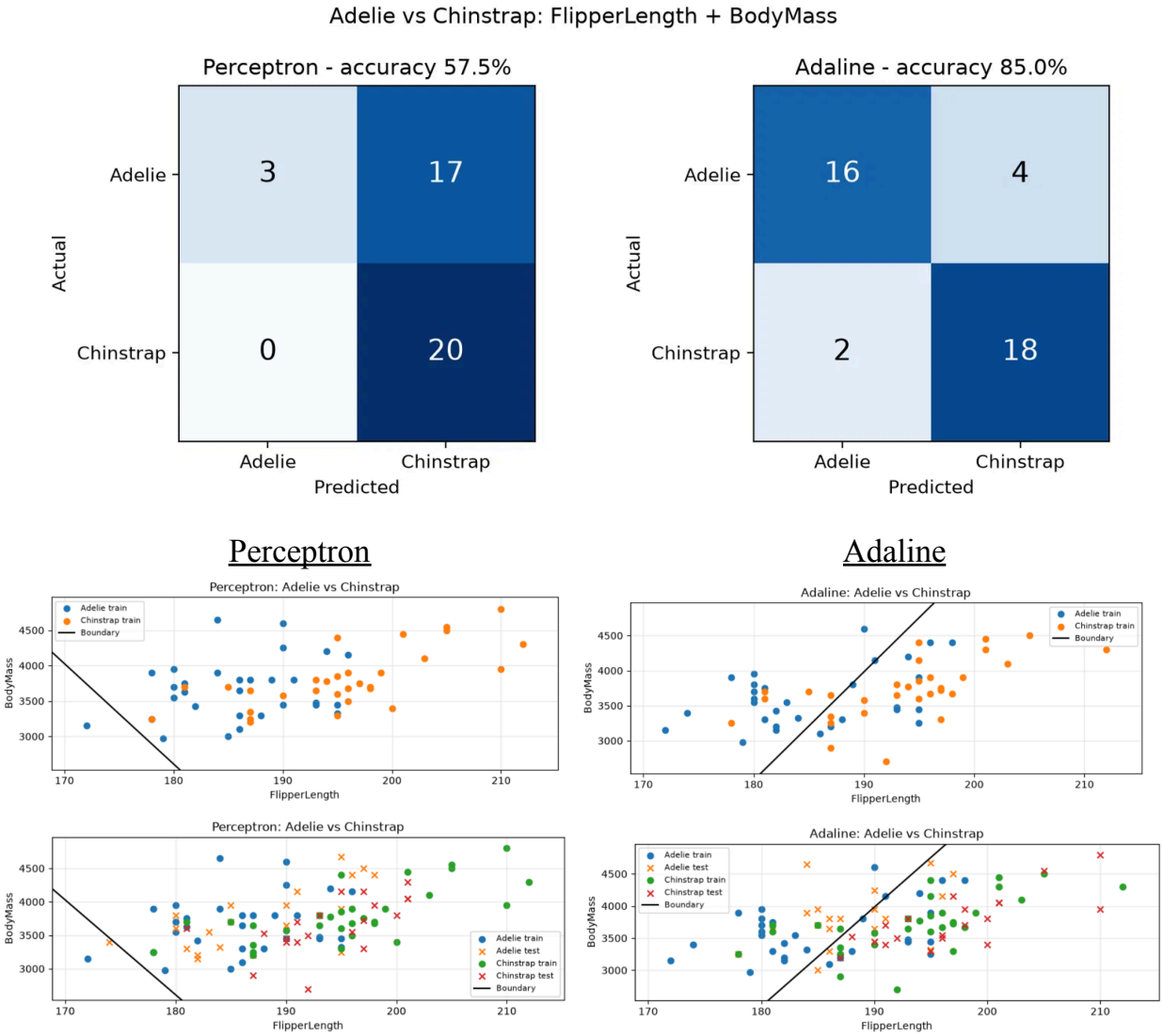
Adaline





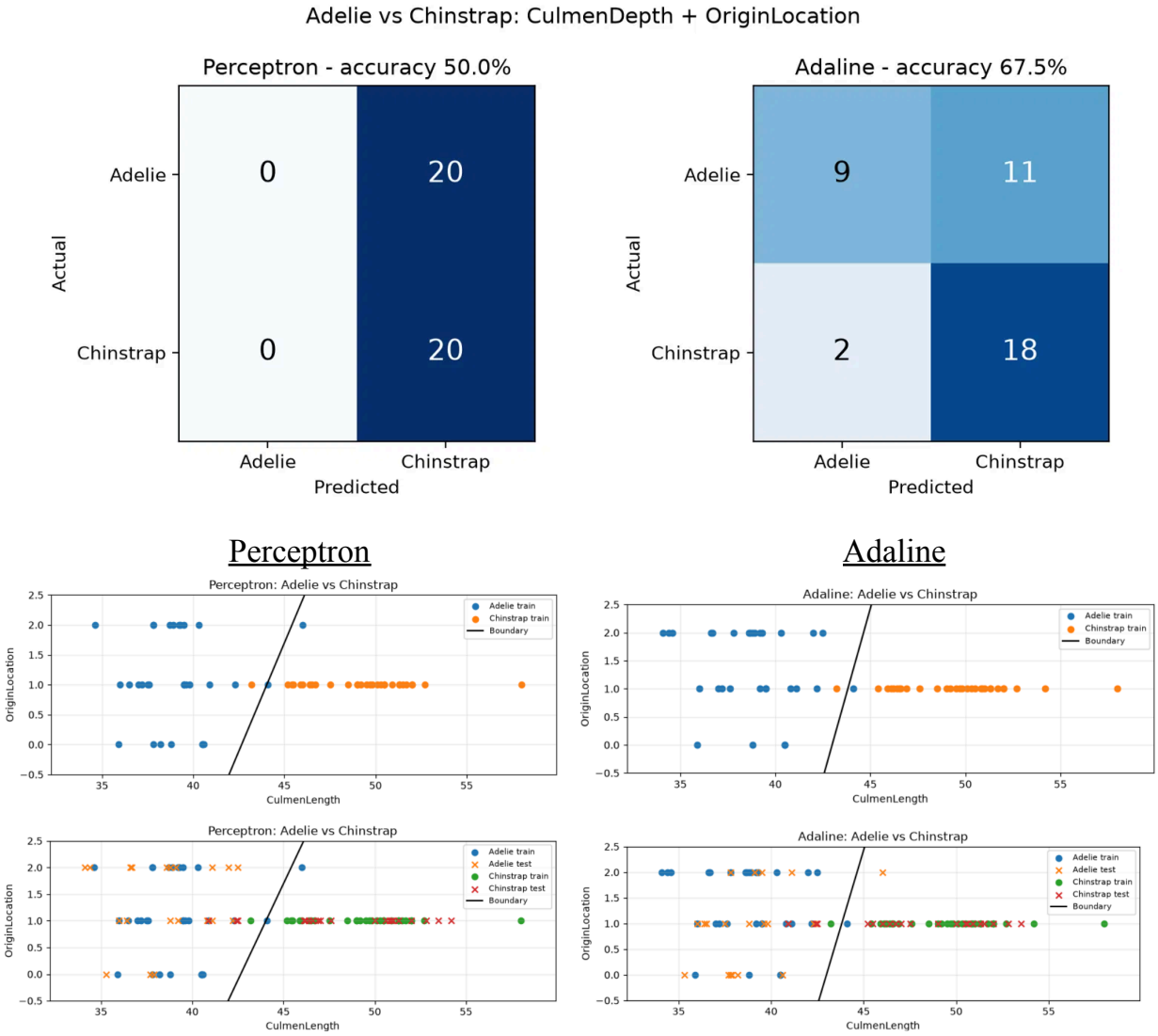
This is a strong example because Adelie and Chinstrap are the hardest species to distinguish. Both algorithms achieve 100% accuracy using CulmenLength and CulmenDepth, as their bill measurements together provide a clear separation. Other feature combinations perform less effectively because flipper length and body mass overlap between the species. This demonstrates that combining both bill measurements is particularly effective for distinguishing Adelie from Chinstrap penguins.

Adelie vs Chinstrap - FlipperLength + BodyMass



Adelie and Chinstrap penguins overlap significantly in flipper length and body mass, making them difficult to separate linearly. The Perceptron achieved only 57.5% accuracy, while Adaline reached 85% by producing a more balanced decision boundary. This suggests that Adaline performs better on overlapping classes because it minimizes squared error, whereas the Perceptron continues updating when the data is not linearly separable. Although exact results vary between splits, Adaline's advantage was consistent.

Adelie vs Chinstrap - CulmenDepth + OriginLocation



This was the weakest feature combination. The Perceptron achieved 50% accuracy by predicting Chinstrap for every sample, while Adaline performed better at 67.5% but still misclassified many Adelie penguins. The poor results are due to overlapping culmen depths and the limitations of encoding OriginLocation as ordered integers, which makes it difficult to distinguish species based on island. One-hot encoding could improve the representation of this categorical feature.

Conclusion & Limitations

Both the Perceptron and Adaline are effective when the classes are linearly separable, and the choice of features then matters more than the choice of algorithm. Adelie vs Gentoo and Chinstrap vs Gentoo were classified almost perfectly with most feature pairs. The hard Adelie vs Chinstrap pair showed the real differences: Adaline's error-minimising training gave a more stable and usually more accurate boundary on overlapping data, while the Perceptron could end with a degenerate boundary because it never converges there. CulmenLength + CulmenDepth was the best feature pair overall.

Limitations

- Each result in Figure 1 and the appendix comes from a single random split with 40 test samples, so one misclassified sample changes accuracy by a lot. Averaging over many random splits would give more reliable comparisons.

- The five detailed examples use a fixed split for a fair comparison, and their numbers differ from the overview run because a different random split was used.

- OriginLocation was encoded as ordered integers, which creates an artificial order between the islands.

Full Results

Species pair	Feature pair	Perceptron	Adaline
Adelie vs Gentoo	CulmenDepth + BodyMass	100.0%	100.0%
Adelie vs Gentoo	CulmenDepth + FlipperLength	100.0%	100.0%
Adelie vs Gentoo	CulmenDepth + OriginLocation	97.5%	100.0%
Adelie vs Gentoo	CulmenLength + BodyMass	92.5%	95.0%
Adelie vs Gentoo	CulmenLength + CulmenDepth	100.0%	100.0%
Adelie vs Gentoo	CulmenLength + FlipperLength	100.0%	100.0%
Adelie vs Gentoo	CulmenLength + OriginLocation	97.5%	100.0%
Adelie vs Gentoo	FlipperLength + BodyMass	100.0%	100.0%
Adelie vs Gentoo	FlipperLength + OriginLocation	100.0%	100.0%
Adelie vs Gentoo	OriginLocation + BodyMass	100.0%	100.0%
Chinstrap vs Gentoo	CulmenDepth + BodyMass	100.0%	100.0%
Chinstrap vs Gentoo	CulmenDepth + FlipperLength	97.5%	100.0%
Chinstrap vs Gentoo	CulmenDepth + OriginLocation	100.0%	100.0%
Chinstrap vs Gentoo	CulmenLength + BodyMass	92.5%	95.0%
Chinstrap vs Gentoo	CulmenLength + CulmenDepth	97.5%	97.5%
Chinstrap vs Gentoo	CulmenLength + FlipperLength	92.5%	92.5%
Chinstrap vs Gentoo	CulmenLength + OriginLocation	100.0%	100.0%
Chinstrap vs Gentoo	FlipperLength + BodyMass	95.0%	97.5%
Chinstrap vs Gentoo	FlipperLength + OriginLocation	100.0%	100.0%
Chinstrap vs Gentoo	OriginLocation + BodyMass	100.0%	100.0%
Adelie vs Chinstrap	CulmenDepth + BodyMass	50.0%	47.5%
Adelie vs Chinstrap	CulmenDepth + FlipperLength	57.5%	75.0%
Adelie vs Chinstrap	CulmenDepth + OriginLocation	45.0%	50.0%
Adelie vs Chinstrap	CulmenLength + BodyMass	92.5%	92.5%
Adelie vs Chinstrap	CulmenLength + CulmenDepth	100.0%	100.0%
Adelie vs Chinstrap	CulmenLength + FlipperLength	82.5%	95.0%
Adelie vs Chinstrap	CulmenLength + OriginLocation	75.0%	95.0%
Adelie vs Chinstrap	FlipperLength + BodyMass	60.0%	77.5%
Adelie vs Chinstrap	FlipperLength + OriginLocation	70.0%	80.0%
Adelie vs Chinstrap	OriginLocation + BodyMass	50.0%	72.5%

Table 4: Complete results table.